

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3.	(a)			${}^2_1\text{H} + {}^3_1\text{H} \rightarrow {}^4_2\text{He} + {}^1_0\text{n}$ H-2 and H-3 correct (1) Neutron symbol correct (1) He-4 correct (1) If all numbers correct but on the RHS of letters then award 2 marks If symbols are correct but LHS and RHS reversed award 2 marks	1	1 1		3		
	(b)	(i)		Any 2 × (1) from: - Need [very] high temperatures / energy - Need high pressures - Difficult to contain - Atoms need enough energy to combine / atoms need to get close enough	2			2		
		(ii)		Long half-life / dangerous (or radioactive) for long periods of time / takes a long time to decay (1) <u>Highly</u> ionising / penetrating (1)	2			2		
				Question 3 total	5	2	0	7	0	0